Water Infrastructure Siting for Employment

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Team Members

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Concise Title of the Problem

Project WISE - Water Infrastructure Siting for Employment

Describe the critical problem you are aiming to solve.

India is facing a surmounting water crisis, which is expected to worsen in the future. The crisis is even more severe in rural India. Several government programmes have the potential to mitigate the problem; however, because of a systemic misalignment among programmes, the problems often exacerbate. India runs one of the largest public employment programmes: The Mahatma Gandhi National Rural Employment Guarantee Act (NREGA). One of the focuses of the NREGA programme is to build water extraction and conservation assets; however, as we illustrate later, the assets often fail to deliver their intended goals. This misalignment stems from a lack of data-driven "co-optimisation" to balance the competing objectives of social welfare (employment) and hydrological impact (groundwater health). As a result, vast resources are spent on assets that are often hydrologically inefficient, socially misplaced, or fail entirely, exacerbating a crisis that affects 600 million people, who face extreme water stress.

Our solution is a decoupled, two-model framework that leverages machine learning and remote sensing to solve this co-optimisation challenge.

1. Model 1 (Priority Map) is a social vulnerability model. It utilises Convolutional Neural Networks (CNNs) to analyse multi-spectral satellite imagery (e.g., nighttime lights, land use) and predict high-resolution poverty and water-need data, thereby overcoming the reliance on outdated census information.
2. Model 2 (Siting Model) is a hybrid learning architecture for hydrological siting. It uses a U-Net (CNN) for deep feature extraction from geospatial data (from sources like CoRE Stack), which identifies complex patterns that manual analysis misses. This rich 'deep feature' vector is then combined with standard terrain data and fed into a suite of robust classifiers such as Support Vector Machines (SVMs), Random Forests (RF), and Gradient Boosted Trees (GBTs) to accurately predict the success of a potential asset site.

To understand the critical importance of this two-model framework, we must first analyse the full scale of the crisis we are addressing.

India is facing the worst water crisis in its history [1]. This is not just a seasonal or cyclical drought pattern, but the result of systematic exploitation and improper resource management. Crucially, this burden falls disproportionately on the rural areas. While urban centres often monopolise the conversation, the distress is most felt in rural India, where 85% of drinking water and over 60% of agriculture rely entirely on groundwater [3]. Unlike cities, where supply chains are diversified, rural livelihoods are tied directly to aquifers that are depleting at alarmingly high rates. According to NITI Aayog, nearly 600 million people face high to extreme water stress, and this number is expected to increase over time [2]. National water demand is projected to be twice the available supply by 2030 [4].

This crisis is not just quantitative; it is also qualitative in nature. India ranks 120th out of 122 countries in the water quality index, as nearly 70% of its water is contaminated. India's main water source is its groundwater. India is the largest user of groundwater and has the largest area under groundwater irrigation globally [5]. However, this resource is in a state of steady decline.

According to a UN-University Institute for Env and Human Security report, “Interconnected Disaster Risk report 2023”, India is going to face groundwater scarcity in 2025, and parts of the Gangetic plains have already gone past the groundwater depletion tipping point [6]. Environmental tipping points are critical thresholds in the Earth’s systems, beyond which abrupt and often irreversible changes occur, leading to profound and sometimes catastrophic shifts in ecosystems, climate patterns and the overall environment. According to the report, 78% of wells in Punjab are considered overexploited, and the north-western region as a whole is predicted to experience critically low groundwater availability by 2025. In regions like Jaipur, this has resulted in 100% over-exploitation across all blocks, with an average annual decline in water levels of 0.50 meters. This depletion is also accompanied by water contamination, with 440 districts reporting excessive nitrate levels in 2023.

However, CGWB data reveals something important. Interventions work when properly implemented. Recharge of groundwater by water conservation structures has increased by 11.36 billion cubic meters since 2017. The percentage of safe groundwater assessment units has increased by 11.8%, and the percentage of over-exploited units has declined by more than 6% [7]. This demonstrates that this crisis is not inevitable. It is merely a problem of efficient planning and scale. The positive impact of existing interventions is being outpaced by depletion, creating an urgent need to accelerate the deployment of correctly sited structures.

India currently has the world's largest public employment program, the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA). It provides at least 100 days of guaranteed wage employment, generating 196.30 crore person-days in FY 2024-2025 alone. It serves as a vital safety net for people living in rural regions of the country [8].

MGNREGA has a dual purpose. It provides a safety net by distributing wages, but it is also a national infrastructure program aimed at the "creation of productive assets of prescribed quality

and durability." The largest category of these assets is "water conservation and water harvesting", creating a bridge to the water crisis [9].

However, in practice, these two objectives are not aligned. In fact, they are sometimes in direct conflict with each other. An asset that would maximise the social objective by creating the most employment could be hydrologically less useful, or even harmful to the hydrological system. Similarly, a planned asset, such as a check dam, situated in an optimal location, may be located in a very remote area where construction might be difficult and benefit a minimal population.

The central problem is therefore one of co-optimisation: how to site assets in a way that efficiently balances these competing objectives.

We can see this misalignment in the performance reports of the scheme. While some MGNREGA projects demonstrate success, such as the recharge of 580 dried-up bore wells in one Andhra Pradesh project, systemic failures are also common [10]. A study by the National Institute of Rural Development and Panchayati Raj (NIRDPR) found more concerning statistics. More than half (62%) of the flood protection works were partially damaged, and one-fourth (26%) canal works were partially damaged. In Gujarat, 10% of Drought Proofing works were partially damaged, and a third of the Water Conservation assets were fully damaged. More than half (76%) of the Drought Proofing works are fully damaged [11].

We are not failing for a lack of will or budget; we are failing because we lack the data and planning to build the right assets in the right places for the right reasons.

This identifies the central problem. We are failing to bridge the gap between India's massive rural employment program and its critical water needs. The problem is a lack of intelligent, data-driven co-optimisation. Without a scalable system to identify optimal locations that balance these competing objectives, MGNREGA will continue to build assets that are hydrologically inefficient or socially misplaced. The challenge is to transform this massive, uncoordinated effort into a precise, effective solution.

Citations

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emand-to-be-twice-the-available-supply-indicating-severe-water-scarcity-report/articleshow/64679218.cms?from=mdr
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11. National Institute of Rural Development and Panchayati Raj. (2021, June 14). *Study on the role of MGNREGA in water conservation and drought mitigation.* http://nirdpr.org.in/nird_docs/rss/rss140621-1.pdf

Why do you think solving this problem is important?

Above all else, this is a matter of improving human lives and protecting the country's ecological system. 600 million people in India are already facing a water crisis. This translates to failed crops, health crises, forced migration, lost livelihoods and even lost lives. Moreover, research indicates that we have already reached an ecological tipping point. The UN report on groundwater depletion in the Gangetic plains signifies that the window to act is closing [1]. Failure to solve this now does not mean a worse problem later; it means an irreversible and devastating disaster. This is no longer a problem of prevention, but rather a last chance for intervention before systemic collapse.

India is already deploying one of the world's largest public employment programs (MGNREGA) and allocating a substantial budget towards it. However, data suggests this investment is not realising its full potential. The structures built may also create a superficial sense of security, even as the underlying crisis continues to accelerate.

We have already noted that the problem is not insurmountable; it is merely a planning and optimisation challenge. Our solution simultaneously enhances the effectiveness of the MGNREGA program while also addressing the water crisis. By creating a data-driven framework

to align these two goals (water and employment), we can develop a scalable solution that benefits many lives across the country.

Citations

1. SANDRP. (2024, January 16). India Groundwater 2023: Reaching Depletion Tipping Point?. https://sandrp.in/2024/01/16/india-groundwater-2023-reaching-depletion-tipping-point/

What is your innovative idea to solve the problem?

As we saw, public administrators face a nearly impossible task: balancing complex hydrological goals with social imperatives. Our innovative idea is to decouple the social and hydrological problems into two sequential models, replacing the all-in-one 'Multi-Objective Optimisation' (MOO) approach.

Model 1, our 'Priority Map,' first identifies the specific geographic areas with the highest social vulnerability (in terms of water and employment needs). To overcome critical data gaps, such as those found in the 2011 Census, this model utilises machine learning and modern satellite imagery to create a high-resolution, up-to-date vulnerability layer.

Model 2, our 'Hydrological Siting Model,' then operates only within these pre-qualified, socially-vulnerable zones to find the optimal site for a specific intervention.

This decoupled framework is innovative because it solves two core problems simultaneously:

1. It's computationally feasible. It drastically reduces the computational load of the hydrological model by constraining the search space before running any complex calculations. MOO methods, including established algorithms like NSGA-II, SMPSO, and EMOACO-I, are computationally demanding by design, requiring many thousands of objective function evaluations.
2. The standard academic approach, a pure Multi-Objective Optimisation (MOO) model, demands a single, unified "objective function". There is no objective, scientific, or ethical exchange rate between one "livelihood-day" and one "litre of aquifer recharge" or one "litre of water yield". Any such function would be an arbitrary political judgment, not a scientific one. By decoupling, we eliminate the arbitrary exchange rate by prioritising human need first.

By separating the problem, we preweight the social component and constrain the search space. If Model 1 filters 90% of the landscape as non-vulnerable, it reduces the computational load of Model 2 by an order of magnitude before any hydrological calculations are run.

The proposed Model 1 is also necessitated by a critical data gap in Indian public administration, characterised by a mismatch in scale and temporality. The only comprehensive, high-resolution

(panchayat-level) socio-economic dataset, the 2011 Census, is temporally obsolete for modern decision-making. Also, all recent, reliable datasets (e.g., the National Family Health Survey 2019-21) are spatially coarse (district-level).
The 2016 Science paper by Jean et al. demonstrates the high feasibility of combining satellite imagery and machine learning to predict poverty. This work established that deep learning models can learn features from both daytime and nighttime satellite imagery that are highly predictive of economic well-being [1]. This forms one of the bases of Model 1.

Our approach builds directly on this, employing a sophisticated and decoupled Multimodal Deep Learning architecture.

Model 1 will be based on Convolutional Neural Networks (CNNs), similar to the cited work, to perform rich feature extraction from multispectral satellite data. This model will learn to correlate these visual features (e.g., road density, building materials, nighttime luminosity) with the 2011 Census data, allowing us to generate high-resolution, modern vulnerability predictions.

Model 2 will leverage a sophisticated hybrid learning architecture, leveraging the strengths of both traditional Machine Learning algorithms and more complicated Deep Learning networks. We will pre-train a Convolutional Neural Network (CNN), such as a U-Net, to detect relevant geospatial patterns in satellite and terrain data. This provides a massive advantage, as the model can identify complex features that manual analysis and human features would miss.

These deep-learned feature vectors will then augment the standard hydrological and terrain data, creating a high-dimensional input for our second-stage classifiers. For this, we will deploy and evaluate a suite of robust supervised models, including Support Vector machines (SVMs), Random Forests, and Gradient Boosted Trees, to perform the final classification and predict site-specific success.

Finally, unlike existing literature that focuses on a single, general-purpose vulnerability map, our approach creates a dynamic system. The features and scores in Model 2 can be re-weighted based on the specific structure being built (e.g., the criteria for a well are different from a check dam), making the targeting far more precise.

Citations

1. Jean, N., Burke, M., Xie, M., Davis, W. M., Lobell, D. B., & Ermon, S. (2016). Combining satellite imagery and machine learning to predict poverty. *Science*, *353*(6301), 790–794. https://doi.org/10.1126/science.aaf7894

How do you plan to implement this idea?

Datasets

The implementation relies on a wide array of geospatial, hydrological, and socio-economic data.

a) Socio-Economic and Job Need Data

1) Panchayat-level NREGA expenditure with equity indicators is a way to estimate the demand for jobs by calculating some deviation. A wider deviation between demand and expenditure would suggest that either adequate demand is not being generated or the expenditure is less than what is required. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=Panchayat%20level%20NREGA%20expenditure%20with%20equity%20indicators)

2) National Family Health Survey provides literacy indicators, poverty indicators, etc. (https://ndap.niti.gov.in/dataset/6822)

3) Census 2011 Data (https://censusindia.gov.in/census.website/data/census-tables)

4) Satellite Imagery, Infrastructure and Vector Data

a) Google Earth Engine (https://earthengine.google.com/)
b) Landsat (https://www.usgs.gov/landsat-missions)
c) Sentinel (https://sentinels.copernicus.eu/)
d) MODIS (https://modis.gsfc.nasa.gov/)
e) CoRE Stack (https://core-stack.org/)

5) Socio-economic variables are available. However, these are outdated. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=Socio%2Deconomic%20variables)

6) Aggregate Development Index estimates present-day poverty levels. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=Aggregate%20Development%20Index)

b) Water Need Data

1) Hydrological layers such as "Net water balance", "recharge: discharge ratio", "average surface water availability during crop seasons", "aquifer type", "stage of groundwater extraction", etc, are available. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=with%20stream%20orders.-,Hydrological%20layers,-%2C%20of%20fortnightly%20time)

2) Climate indicators such as "Drought intensity" and "drought frequency". (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=in%20water%20balance-,Climate%20indicators,-%3A%20Drought%20intensity%20and)

3) Land use and land cover classification. A high percentage of cropland increases the SEVI (Socioeconomic Vulnerability Index) score. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=Land%20use%20and%20land%20cover%20classification)

4) Also need to consider Terrain Classification as you can't build a well anywhere. (https://core-stack.org/core-stack-geospatial-layers-and-datasets/#:~:text=Terrain%20classification)

5) ISRO Bhuvan provides groundwater yield data. (https://bhuvan-app1.nrsc.gov.in/bhuvan2d/bhuvan/bhuvan2d.php)

6) The Central Groundwater Board has data for groundwater stress/over-exploitation at the district levels. It also maintains records of individual wells monitored by both the central and state governments, including well depth and groundwater levels. (https://gwdata.cgwb.gov.in/)

7) INDIA WRIS API Catalogue contains data on Ground Water Measurement Stations and more information like rainfall, soil moisture, etc. (https://indiawris.gov.in/swagger-ui/index.html)

8) Groundwater Quality Report 2023 contains data on groundwater quality and contamination throughout various stations in India. (https://cgwb.gov.in/sites/default/files/inline-files/final_nhs-wq_pre_2023_compressed.pdf)

9) State-wise data on groundwater contamination. (https://www.data.gov.in/resource/stateut-wise-details-nitrate-fluoride-arsenic-and-uranium-contamination-ground-water)

10) Data on groundwater quality throughout various stations in India. (https://indiawris.gov.in/wris/#/GWQuality)

Model 1 (Priority Map)

To build this Priority Map, we must first address a core data problem. While many of the datasets listed earlier are updated and recent, the main socio-economic data from the 2011 Census is outdated and unreliable for guiding modern MGNREGA allocations.

We need to identify which villages are currently the most vulnerable and have the highest needs for water and employment. However, due to the postponement of the 2021 Census, the 2011 Census remains the only comprehensive, high-resolution (panchayat-level) source for this. Using 2011 data for 2025 decisions is highly unreliable, as villages undergo significant changes over a 14-year period.

1) To bridge this gap, one may implement alternative datasets, such as:
    - NITI Aayog MPI, sourced from the National Family Health Survey, is for the period 2019-21. However, this dataset is reliable only on the district level, not at the panchayat level.
    - The Household Consumption Expenditure Survey 2023-24 is also very recent, but it was conducted only at the state and national levels.
    - The Periodic Labour Force Survey from 2023-24 is a sample survey and is only reliable at the state and national levels.
    - National Accounts Statistics is also available at the state and national levels.

2) The limitation of these alternatives is their scale. A district is a vast, diverse area, often comprising both a wealthy, urban district headquarters and 200 remote, off-grid villages. A district-level dataset (such as NFHS) averages these differences, making it less useful for our goals. It hides the extreme poverty in the remote villages by averaging it with the wealth of the city. We need to find the specific hotspots of vulnerability within the district.

3) This leaves us with a 14-year data gap, which is large enough to render the data almost obsolete in our case. Therefore, we propose building a model.
    - The model will be trained on (2011 Satellite Imagery + 2011 Terrain) -> (2011 Census Data). It can then be used to predict 2025 socio-economic details using 2025 Satellite Imagery and Terrain data.
    - This satellite data contains many features that the model can correlate with the census data, such as nighttime lights, building materials, roads, land use, and settlement patterns.

4) Thus, we propose a two-phase plan:

    a) Phase 1: Identify Vulnerable Districts.
    i) We will filter the 800 districts in India to a more manageable number, using the following three layers:
        - Layer 1 (Socio-Economic Data) - NFHS (2019-21) district-level data will be used to find the districts with the worst scores for poverty, literacy, and household assets.
        - Layer 2 (Job Need Data) - MGNREGA data will be utilised to identify districts with the highest unmet job demand. However, since this only provides data on employment under this particular scheme, data from other sources (such as the National Data and Analytics Division Portal and State Government Data) should also be considered.
        - Layer 3 (Water Need Data) - Refer to the datasets previously listed.

    ii) After collecting this data, we will then normalise the data, and find a vulnerability score for each district for each layer. Finally, we will calculate a subjective, weighted final vulnerability score by consulting experts and examining correlations in the data.

    b) Phase 2: Create a high-resolution vulnerability map only within the identified districts.

    i) Layer 1 (Socio-Economic Data): We will run the (2011 Census -> 2011 Satellite) model to predict 2025 vulnerability using 2025 satellite imagery. This approach saves computing power and time.
    For validation, we can average our predictions up to the district level and compare them against the NFHS data to assess the model's accuracy.
    ii) Layer 2 (Job Need Data): Again, we will use the MGNREGA data, State Government Data and other relevant data provided in Core-Stack.
    iii) Layer 3 (Water Need Data): Refer to the datasets previously listed.

6) We can then normalise these three panchayat-level layers and combine them using the same weighted formula as before.

7) Note that all three layers have to be customised and take different things into account for different structures. For example:

i) Wells

a) In Phase 1, NFHS-5 indicators such as "percentage of households without an improved drinking-water source" and "percentage of children with diarrheal disease" can be upweighted. Literacy and poverty are secondary. The CGWB data on groundwater quality can be upweighted and filtered by feasibility, excluding over-exploited areas.

b) In Phase 2, additional data, such as geospatial information on distance to existing piped water lines or public standposts, can be incorporated. ISRO Bhuvan data can also be used.

ii) Check Dams

a) In Phase 1: Agricultural employment and other agricultural indicators, as well as high rainfall variability, high drought frequency, and "Over-Exploited" groundwater status indicators, can be upweighted.

b) In Phase 2, a High percentage of rain-fed cropland should be accounted for. Areas with moderate slopes and high runoff, along with the "recharge: discharge ratio," can be used to ensure the ground is suitable for infiltration.

Model 2 (Siting Model)

With the socially vulnerable areas identified by Model 1, Model 2 is used to find the optimal physical site for an asset. To achieve this, we need to create distinct workflows for different structures, as they are fundamentally distinct.

Check Dams and Recharge Structures

1) The Ground Truth:
To train a model that predicts good sites, we first need to define "ground truth" by identifying existing successful and unsuccessful dams.

i) First, the coordinates of the existing check dams are required. This can be done by:

a) Utilising MGNREGA data on check dams that were built under the scheme, as many of the assets are also geotagged.

b) Training a simple U-Net on high-resolution Satellite Data to detect and map existing check dams.

ii) For each dam, a "before and after" analysis will be performed using historic satellite data such as Landsat.

iii) A Digital Elevation Model (DEM) will be used to automatically delineate the small watershed and downstream "Area of Influence" (AOI) for each dam.

iv) The average NDVI (Normalised Difference Vegetation Index) will be calculated for 3 years before and 3 years after the dam was built to find the change in NDVI. This value serves as the success score.

v) "Bad" sites are also needed for the negative samples. The DEM can be used to identify all potential stream-order locations that are hydrologically similar to the good sites.

2) Feature Engineering:
Once we have identified the ground truth sites, we will develop a feature vector for every spot on the map, incorporating features such as terrain, hydrology, soil data, geology data, land use, and infrastructure.

3) Model Training:
We will deploy a hybrid, two-stage learning architecture to maximise predictive accuracy. The final site-suitability classification will be handled by a suite of robust supervised models, including Support Vector Machines (SVMs), Random Forests (RF), and Gradient Boosted Trees (GBTs).

These classifiers will be trained on a hybrid feature set:

- Engineered Features: Our manually-derived hydrological, geological, and terrain features.
- Deep Features: This is where we leverage transfer learning. The previously trained U-Net model will act as a powerful, automated feature extractor. By passing the geospatial data for a potential site through the pre-trained U-Net, we will extract its high-level, latent-space representations. These "deep features," which capture complex spatial patterns a human would miss, will be concatenated with our engineered features.

This two-stage approach provides the best of both worlds: the proven classification power of GBTs and SVMs, augmented by the rich, nuanced feature-extraction capabilities of Deep Learning.

Wells

1) The Ground Truth
For wells, the ground truth data is also based on the performance of existing structures.

i) We will use the CGWB, MGNREGA data, and state government records to find well depth, well yield, and water level, along with their coordinates.

2) Feature Engineering:
The feature set for this model will include geology, hydrology, terrain, long-term NDVI, etc.

3) Model Training: As with the check dam model, we will train and evaluate Support Vector Machines (SVMs), Random Forests, and Gradient Boosted Trees to predict well success.

Since wells are point data rather than the linear/area features of a check dam (for which a U-Net is ideal), we will test two advanced architectures:

1. A Patch-Based CNN: We will train a Convolutional Neural Network (CNN) that analyses a 1km x 1km "patch" of multi-layer data (satellite imagery, geology, DEM, etc.) centred on each well. This allows the model to learn the optimal spatial combination of features associated with high-yield wells.
2. A Hybrid Model: Similar to the check dam workflow, we will use this patch-based CNN as a feature extractor. We will then feed its learned feature vector—alongside our core-engineered features (e.g., specific aquifer type, local hydrogeology)—into our SVMs and GBTs.

This allows us to explore whether an end-to-end DL model or a hybrid ML-DL architecture provides the most robust predictions for groundwater potential.

Have you done any prior research related to the problem?

While our team has not conducted prior research specifically within the rural development or hydrological siting domain, we are approaching this problem with extensive, hands-on expertise in the core machine learning and deep learning methodologies that this solution requires.

- Deep Learning & Hybrid Architectures: Our team has significant experience developing and implementing hybrid deep learning models. Key projects include a hybrid ConvLSTM network for Time Series Anomaly Detection in WBANs. This work on complex sequential data and hybrid models is directly relevant to building the proposed satellite-imagery-based CNN (Model 1) and the hybrid ML-DL architectures (Model 2).
- Classical Machine Learning: We have practical, hands-on experience using the exact classifiers proposed for Model 2.
- Broad ML/DL Proficiency: Collectively, our team is highly familiar with all the classic supervised algorithms central to this project, including Random Forests (RF), Gradient Boosted Trees (GBTs), SVMs and K-means clustering, having applied them to various problems.

How do you think your solutions will benefit rural communities?

Our solution provides direct, tangible benefits to rural communities by transforming the MGNREGA program from an often uncoordinated and wasteful effort into a precise and effective solution for both water scarcity and livelihood security.
The core problem our solution addresses is the "direct conflict" between MGNREGA's two primary objectives: providing immediate employment and creating durable water conservation assets. This misalignment often results in assets being built in locations that are hydrologically

useless or socially misplaced, leading to systemic waste and high failure rates. Studies have found that over 76% of drought-proofing works are "fully damaged" [4].

Our decoupled, two-model approach corrects this fundamental flaw, ensuring that public investment yields lasting benefits for the communities that need them most.

Model 1: Prioritising People

Our "Priority Map" first identifies the geographic areas of highest social vulnerability. This model targets the specific communities with the most urgent combined need for both water and employment.

1. Benefit: This model overcomes a critical data gap, which was the reliance on obsolete 2011 census data, by using machine learning and modern satellite imagery to predict *current*, high-resolution, panchayat-level vulnerability.
2. Impact: This ensures that interventions are directed to the right people, stopping the failure of building assets in locations that do not address the most pressing social needs.

Model 2: Ensuring Effectiveness

Operating *only* within these socially prioritised zones, our "Siting Model" then identifies the most hydrologically optimal locations for water assets, such as check dams and wells.

1. Benefit: This model is trained on the real-world performance of existing assets, using data-driven analysis to find new locations with the highest probability of success (e.g., maximising aquifer recharge or well yield).
2. Impact: This ensures that the assets built are effective, durable, and create a sustainable water supply, directly tackling the crisis facing 600 million people [1] and declining groundwater levels [2].

This two-step framework provides a dual benefit for rural communities:

1. Immediate Livelihood: It provides a vital and reliable social safety net through guaranteed MGNREGA employment [3].
2. Long-Term Resilience: For the first time, it guarantees that this employment creates productive, durable water assets.

With this alignment of social and hydrological objectives, our solution promotes long-term water security, a key factor in preventing crop failures, health crises, and forced migration - thereby delivering on the program's intended promise of a stable livelihood and a resilient environment.

Citations

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[2].India Groundwater 2023: Reaching Depletion Tipping Point? – SANDRP

[3].Year End Review 2024: Achievement of the Department of Rural Development

[4].A Study of Productivity and Sustainability of MGNREGS Assets in Selected Districts from All the States in India

Please write quarterly milestones for your project.

Quarter 1: Data Acquisition and Initial Model 1 Prototyping

1) Secure, clean, and preprocess all required datasets, including socio-economic (NFHS, NREGA, 2011 Census), water (CGWB, WRIS, CoRE Stack), and geospatial (Landsat, Sentinel) data.
2) Prototype the socio-economic CNN model on (2011 Satellite Imagery + 2011 Terrain) to predict (2011 Census Data) and refine data collection, weighting and engineering based on the results.
3) Evaluate the feasibility and performance of the prototyped Model 1, and repeat the process to achieve the best results.

Quarter 2: Model 1 (Priority Map) Development and Validation

1) Train the core socio-economic CNN model on (2011 Satellite Imagery + 2011 Terrain) to predict (2011 Census Data).
2) Apply the trained model to 2025 satellite data to generate high-resolution socio-economic predictions; validate these predictions by aggregating them to the district level and comparing them against NFHS data.
3) Integrate the predicted socio-economic layer with high-resolution job need (MGNREGA) and water need (hydrological) data to produce the final, validated, panchayat-level "Priority Map".
4) Begin data collection, cleaning, and preprocessing for Model 2.

Quarter 3: Model 2 (Check Dam Siting) Development and Training

1) Generate the "Ground Truth" dataset for check dams by mapping existing structures (using MGNREGA data or a U-Net) and calculating "success scores" via historical NDVI analysis.
2) Develop the hybrid learning architecture: complete manual feature engineering and perform deep feature extraction using the U-Net from geospatial data.
3) Train and evaluate the final check dam siting classifiers (SVMs, RF, GBTs) using the combined feature sets.

Quarter 4: Model 2 (Well Siting) Development and Final System Integration

1) Generate the "Ground Truth" dataset for wells using CGWB and state government data (well yield, depth, water level).
2) Train, test, and evaluate both proposed architectures for well siting: the Patch-Based CNN and the Hybrid ML-DL model.
3) Select the best-performing models for both wells and check dams and integrate them with Model 1 (Priority Map) to create the final decision framework.
4) Final result dissemination and analysis.
5) Transferring the technology to the stakeholders with a proper framework for use
6) Publishing the model architectures, methodologies and results